SQL Team Progress Report: Week 1 Onboarding

Project: Ride-Share Demand Analysis (`ride-share-demand-analysis`)

SQL Members: Gideon, Rolland, Belinda.

Date: March 31, 2026

1. Environment Initialization (BigQuery)

- **Project Provisioning:** Successfully created the Google Cloud Project and linked it to the `bigquery-public-data` repository.
- **Dataset Identification:** Verified the source table: `bigquery-public-data.chicago_taxi_trips.taxi_trips`.
- **Schema Audit:** Conducted a full review of the 23 available columns to identify primary keys (`unique_key`) and metric columns (`fare`, `trip_start_timestamp`, `trip_seconds`).

2. Security & Budgetary Controls

- **Billing Activation:** Successfully transitioned the project from "Sandbox" to a full account using the $300 student credit.
- **Risk Mitigation:** * Implemented a **$10 Monthly Budget Alert** to monitor query costs.
    - Enforced a **10 GB per-query scan limit** policy for the team to ensure efficient credit usage.
- **IAM Access:** Granted permissions to all team members, ensuring SQL workspace visibility and Dataform repository access.

3. SQL Baseline & Performance Optimization

We have established a "Safe Query Template" for all Week 1 exploration. This template prevents accidental full-table scans of the 200M+ row dataset.

Standardized Baseline Query:

```SQL
-- Purpose: Initial Data Validation for 2023
-- Strategy: Use 'WHERE' for Partition Filtering and 'LIMIT' for Preview
SELECT
 trip_start_timestamp,
 fare,
 tips,
 trip_total,
```

```sql
 trip_seconds,
 pickup_community_area
FROM `bigquery-public-data.chicago_taxi_trips.taxi_trips`
WHERE EXTRACT(YEAR FROM trip_start_timestamp) = 2023
  AND EXTRACT(MONTH FROM trip_start_timestamp) = 1
LIMIT 1000;
```

4. Week 1 Checklist Status

Task	Status	Notes
BigQuery Project Setup	☑ **Complete**	ID: `ride-share-demand-analysis`
Table Exploration	☑ **Complete**	Schema reviewed for null values in `fare`.
IAM Permissions	☑ **Complete**	Access granted
Billing/Credits Alert	☑ **Complete**	$10 safety alert active.
Initial SQL Script	☑ **Complete**	Saved as `01_january_2023_preview`.

Next Objectives (Week 2):

- Execute aggregate queries for annual 2023 totals.
- Analyze trip distribution by `community_area`.
- Identify peak demand hours via `timestamp` extraction.